

Failure of fiberwise log-plurisubharmonicity for classical p -Bergman kernels

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Abstract

We show that fiberwise log-plurisubharmonicity of the classical p -Bergman kernel fails for every $p > 2$. For $\alpha = 1 + p/2$, the pseudoconvex Hartogs family defined by $|\eta|^2(|z|^2 + |1 + wz|^2)^\alpha < 1$ has the explicit fiberwise kernel $K_{X_w,p}(0, 0) = \frac{p}{2\pi^2}(1 + |w|^2)^{1-p/2}$. Its logarithm has strictly negative Levi form along the fixed holomorphic section $(0, 0, w)$. Polynomial decay forces the associated weighted entire-function space to consist only of constants, and an elementary change of variables evaluates the extremal integral. Exhaustion yields bounded pseudoconvex counterexamples for every $p > 2$. Together with the known positivity for $0 < p \leq 2$, this identifies the sharp exponent range.

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1 Introduction

For a domain $D \subset \mathbb{C}^n$ and $p > 0$, set

$$A^p(D) := \left\{ f \in \mathcal{O}(D) : \int_D |f|^p dV < \infty \right\}, \quad K_{D,p}(a) := \sup_{0 \neq f \in A^p(D)} \frac{|f(a)|^p}{\int_D |f|^p dV},$$

where dV denotes Euclidean Lebesgue measure. Whenever evaluation at a is nonzero on $A^p(D)$, this is equivalently

$$K_{D,p}(a) = \left(\inf \left\{ \int_D |f|^p dV : f \in A^p(D), f(a) = 1 \right\} \right)^{-1}.$$

The variation theory of Bergman kernels originates in the Hilbert-space case $p = 2$. Maitani–Yamaguchi [8] proved plurisubharmonic variation of the logarithm of the Bergman kernel for families of open Riemann surfaces. Berndtsson [1] established the corresponding result for general pseudoconvex

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families and developed its vector-bundle interpretation through positivity of direct images [2]; see also Berndtsson–Păun [4] for relative Bergman kernels and pseudoeffectivity.

The L^2 extension theory provides a closely related route to Bergman-kernel estimates. Starting from the Ohsawa–Takegoshi extension theorem [9], sharp forms and applications were obtained by Blocki [6] and Guan–Zhou [7]. Berndtsson–Lempert [3] gave a proof with sharp estimates using positivity and variation of domains.

For pseudoconvex families with bounded fibers, fiberwise log-plurisubharmonicity of the classical p -Bergman kernel, and more generally of the kernel associated with a functional ξ , holds for $0 < p \leq 2$; see Bao–Guan–Sun [5, Section 6.1]. The following theorem shows that this exponent range is sharp, already for the classical kernel and fibers of complex dimension two.

THEOREM 1.1. *Let $p > 2$, put $\alpha = 1 + p/2$, and define*

$$S(z, w) := |z|^2 + |1 + wz|^2. \quad (1.1)$$

The domain

$$\mathcal{X}_p := \{(z, \eta, w) \in \mathbb{C}^2 \times \mathbb{D} : |\eta|^2 S(z, w)^\alpha < 1\} \quad (1.2)$$

is pseudoconvex. For each $w \in \mathbb{D}$, let

$$X_{p,w} := \{(z, \eta) \in \mathbb{C}^2 : (z, \eta, w) \in \mathcal{X}_p\}.$$

Then

$$K_{X_{p,w},p}(0, 0) = \frac{p}{2\pi^2} (1 + |w|^2)^{1-p/2}, \quad (1.3)$$

and consequently

$$\frac{\partial^2}{\partial w \partial \bar{w}} \log K_{X_{p,w},p}(0, 0) = \frac{1 - p/2}{(1 + |w|^2)^2} < 0. \quad (1.4)$$

Moreover, for every sufficiently large R , the bounded pseudoconvex domain $\mathcal{X}_p \cap \{|z| < R\}$ also fails fiberwise log-plurisubharmonicity of the classical p -Bergman kernel.

For example, when $p = 3$, one has $\alpha = 5/2$ and the Levi form in (1.4) equals $-1/2$ at $w = 0$. The proof first identifies the normalized extremal function as the constant function, then evaluates its norm exactly. The bounded counterexamples follow from a strict violation of the submean inequality and monotone convergence under exhaustion.

2 The Hartogs family and the extremal function

Throughout the proof, fix $p > 2$ and $\alpha = 1 + p/2$. The holomorphic functions z and $1 + wz$ have no common zero. Hence

$$\Phi(z, w) := \alpha \log S(z, w)$$

is smooth and plurisubharmonic on $\mathbb{C} \times \mathbb{D}$. The complete Hartogs domain $\mathcal{X}_p = \{|\eta|^2 < e^{-\Phi(z,w)}\}$ is therefore pseudoconvex.

For later use, put $B = 1 + |w|^2$. Completing the square gives

$$S(z, w) = \frac{1 + |Bz + \bar{w}|^2}{B}. \quad (2.1)$$

In particular, $S(z, w) \geq B^{-1} > 1/2$ for $w \in \mathbb{D}$. Also, uniformly on $\mathbb{C} \times \mathbb{D}$,

$$\frac{1 + |z|^2}{3} \leq S(z, w) \leq 3(1 + |z|^2). \quad (2.2)$$

For the lower bound, use $1 \leq 2|1 + wz|^2 + 2|z|^2$; the upper bound follows from $|1 + wz|^2 \leq 2 + 2|z|^2$. Thus

$$I_\alpha(w) := \int_{\mathbb{C}} S(z, w)^{-\alpha} dA(z) < \infty, \quad (2.3)$$

where dA is planar Lebesgue measure, since $\alpha > 1$.

LEMMA 2.1. *For every $w \in \mathbb{D}$, the space*

$$\left\{ f \in \mathcal{O}(\mathbb{C}) : \int_{\mathbb{C}} |f(z)|^p S(z, w)^{-\alpha} dA(z) < \infty \right\}$$

consists exactly of the constant functions.

Proof. Constants are integrable by (2.3). Conversely, write an integrable entire function as $f(z) = \sum_{k \geq 0} a_k z^k$. Cauchy's coefficient formula and Hölder's inequality give

$$|a_k|^p r^{kp} \leq \frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})|^p d\theta.$$

By the upper bound in (2.2), $S^{-\alpha} \geq 3^{-\alpha}(1 + |z|^2)^{-\alpha}$. Multiplying the preceding inequality by $r(1 + r^2)^{-\alpha}$ and integrating shows that $a_k \neq 0$ requires

$$\int_1^\infty r^{kp+1} (1 + r^2)^{-\alpha} dr < \infty.$$

Since $2\alpha = p + 2$, the integrand is asymptotic at infinity to $r^{(k-1)p-1}$. For every $k \geq 1$ the integral diverges. In particular, $k = 1$ gives logarithmic divergence at the endpoint $2\alpha = p + 2$. Therefore $a_k = 0$ for all $k \geq 1$. $\blacksquare$

PROPOSITION 2.2. *For every $w \in \mathbb{D}$, the unique normalized extremal function at $(0, 0)$ is $F \equiv 1$, and*

$$K_{X_{p,w,p}}(0, 0) = \frac{1}{\pi I_\alpha(w)}. \quad (2.4)$$

Proof. Let $F \in A^p(X_{p,w})$ satisfy $F(0, 0) = 1$ and put $f(z) = F(z, 0)$. Since $(z, 0) \in X_{p,w}$ for all $z \in \mathbb{C}$, the function f is entire. Applying the submean inequality in the η -disk and then Tonelli's theorem gives

$$\int_{X_{p,w}} |F|^p dV \geq \pi \int_{\mathbb{C}} |f(z)|^p S(z, w)^{-\alpha} dA(z). \quad (2.5)$$

The right side is finite, so Lemma 2.1 and $f(0) = 1$ imply $f \equiv 1$. Hence every admissible F has p -th power norm at least $\pi I_\alpha(w)$. The constant function $F \equiv 1$ belongs to $A^p(X_{p,w})$, and its p -th power norm is exactly $\pi I_\alpha(w)$. Thus the infimum is attained and (2.4) follows. Uniqueness follows from strict convexity of $|\cdot|^p$ for $p > 1$ on the affine space of normalized holomorphic functions. $\blacksquare$

3 Exact kernel formula and negative curvature

For a fixed w , use (2.1) and the change of variables $\zeta = Bz + \bar{w}$. Its real Jacobian is B^2 , so

$$\begin{aligned} I_\alpha(w) &= B^{\alpha-2} \int_{\mathbb{C}} (1 + |\zeta|^2)^{-\alpha} dA(\zeta) \\ &= 2\pi B^{\alpha-2} \int_0^\infty r(1 + r^2)^{-\alpha} dr \\ &= \frac{\pi}{\alpha-1} (1 + |w|^2)^{\alpha-2}. \end{aligned} \quad (3.1)$$

This substitution is performed separately on each fiber; no holomorphic dependence of the change of variables on w is needed. Combining Proposition 2.2 with (3.1) yields

$$K_{X_{p,w},p}(0,0) = \frac{\alpha-1}{\pi^2}(1+|w|^2)^{2-\alpha} = \frac{p}{2\pi^2}(1+|w|^2)^{1-p/2}.$$

Since

$$\frac{\partial^2}{\partial w \partial \bar{w}} \log(1+|w|^2) = \frac{1}{(1+|w|^2)^2},$$

formula (1.4) follows. The map $w \mapsto (0,0,w)$ is a holomorphic section of $\mathcal{X}_p$. If the logarithm of the fiberwise kernel were plurisubharmonic on $\mathcal{X}_p$, its restriction to this section would be subharmonic, contradicting (1.4).

REMARK 3.1. *The endpoint choice $\alpha = 1 + p/2$ is not essential. For any $2 < \alpha \leq 1 + p/2$, the same coefficient argument excludes every nonconstant entire function, since integrability requires $kp < 2\alpha - 2 \leq p$. The kernel is then $(\alpha - 1)\pi^{-2}(1+|w|^2)^{2-\alpha}$, with Levi form $(2 - \alpha)/(1+|w|^2)^2$. Such an α exists precisely when $p > 2$.*

4 Bounded pseudoconvex counterexamples

For $R > 0$ define

$$\mathcal{X}_p^{(R)} := \mathcal{X}_p \cap \{|z| < R\}, \quad X_{p,w}^{(R)} := X_{p,w} \cap \{|z| < R\}.$$

These are pseudoconvex domains. The total space $\mathcal{X}_p^{(R)}$ is bounded: $|w| < 1$, $|z| < R$, and (2.1) gives $|\eta|^2 < S^{-\alpha} \leq (1+|w|^2)^\alpha < 2^\alpha$.

LEMMA 4.1. *For every fixed $w \in \mathbb{D}$,*

$$K_{X_{p,w},p}^{(R)}(0,0) \searrow K_{X_{p,w},p}(0,0) \quad (R \rightarrow \infty).$$

Proof. Write m_R for the infimum of $\int_{X_{p,w}^{(R)}} |F|^p dV$ under the constraint $F(0,0) = 1$, and write m_∞ for the corresponding infimum on $X_{p,w}$. Restriction gives $0 < m_R \leq m_{R'} \leq m_\infty$ for $R < R'$. Positivity follows from the submean inequality on a ball about $(0,0)$, and finiteness follows from the admissible constant function. Let $m_* := \lim_{R \rightarrow \infty} m_R$.

Choose $R_j \rightarrow \infty$ and normalized holomorphic functions F_j on $X_{p,w}^{(R_j)}$ such that

$$\int_{X_{p,w}^{(R_j)}} |F_j|^p dV \leq m_{R_j} + j^{-1}.$$

The local submean inequality bounds F_j uniformly on each compact subset of $X_{p,w}$. A diagonal normal-family argument gives a subsequence converging locally uniformly to $F \in \mathcal{O}(X_{p,w})$ with $F(0,0) = 1$. Applying Fatou's lemma on each fixed truncation and then letting the truncation increase yields

$$\int_{X_{p,w}} |F|^p dV \leq m_*.$$

Consequently $m_\infty \leq m_*$. The reverse inequality was already established, so $m_R \nearrow m_\infty$. Taking reciprocals proves the claim. This argument does not require prior attainment on the truncated domains. $\blacksquare$

COROLLARY 4.2. *For every $p > 2$ and every sufficiently large R , the function*

$$(z, \eta, w) \mapsto \log K_{X_{p,w,p}^{(R)}}(z, \eta)$$

is not plurisubharmonic on $\mathcal{X}_p^{(R)}$.

Proof. Put $u_R(w) = \log K_{X_{p,w,p}^{(R)}}(0, 0)$ and $u_\infty(w) = \log K_{X_{p,w,p}}(0, 0)$. By Lemma 4.1, $u_R \searrow u_\infty$ pointwise. The unitary map $(z, \eta) \mapsto (e^{-i\theta}z, \eta)$ identifies the fiber at w with the fiber at $e^{i\theta}w$, preserves $|z| < R$, and fixes $(0, 0)$. Therefore u_R is radial in w .

Fix $r = 1/2$. Formula (1.3) gives the strict gap

$$u_\infty(0) - u_\infty(r) = \left(\frac{p}{2} - 1\right) \log \frac{5}{4} > 0.$$

Pointwise convergence at 0 and r shows that $u_R(0) > u_R(r)$ for every sufficiently large R . If the fiberwise logarithmic kernel on $\mathcal{X}_p^{(R)}$ were plurisubharmonic, u_R would be subharmonic. Its radially would then give

$$u_R(0) \leq \frac{1}{2\pi} \int_0^{2\pi} u_R(re^{i\theta}) d\theta = u_R(r),$$

a contradiction. ■

This completes the proof of Theorem 1.1. The strict violation is obtained from kernel values at two parameter radii; convergence of Levi forms under exhaustion is not asserted or required. Together with the positivity result for $0 < p \leq 2$ in [5, Section 6.1], the corollary shows that $0 < p \leq 2$ is the exact exponent range for universal fiberwise log-plurisubharmonicity on bounded pseudoconvex families.

AI declaration and verification status

The original construction and proof were obtained with GPT-5.6 Sol (high). The extension to every $p > 2$, the explicit kernel formula, and the revised exposition were developed with OpenAI Codex (GPT-6). **Verification by the author is in progress.** The revised manuscript has not yet been fully independently verified; the verification status does not indicate peer review.

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